

Homework 3

EE 999 — Introduction to Control Systems

Patrick Gordon • September 27, 2026

Due Wednesday, September 23, 2026

Problem 1 — Matlab

Code: ./hw_03_01.m

```
1 if exist('OCTAVE_VERSION', 'builtin')
2     pkg load control;
3 end
4
5 % Note these values were chosen to accentuate the differences
6 Kp = 1000;
7 Ki = 1000;
8 Kd = 1000;
9
10 %=====
11
12 m = 1000;
13 b = 50;
14 r = 1;
15
16 s = tf('s');
17 P_cruise = 1/(m*s + b);
18 t = 0:0.1:20;
19
20 %=====
21
22 C = pid(Kp);
23 T_P = feedback(C*P_cruise, 1);
24
25 %=====
26
27 C = pid(Kp, Ki);
28
29 T_PI = feedback(C*P_cruise, 1);
30
31 %=====
32
33 C = pid(Kp, Ki, Kd);
34
35 T_PID = feedback(C*P_cruise, 1);
36
37 %=====
38
39 figure
40 step(r*T_P, 'b', r*T_PI, 'r', r*T_PID, 'g', t)
41 axis([0 20 0 1.5*r])
42 legend('P', 'PI', 'PID', 'Location', 'southeast')
```

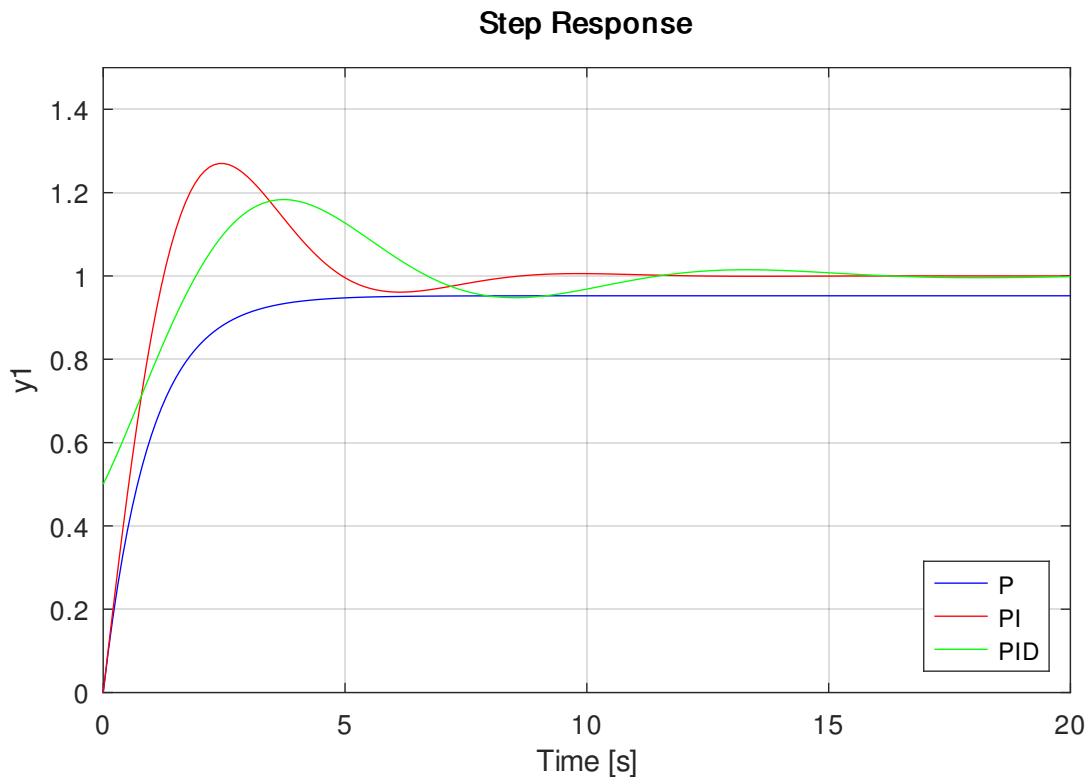


Figure 1: Step response of the cruise control system under P, PI, and PID control.

Problem 2.A — Realistic Op Amp

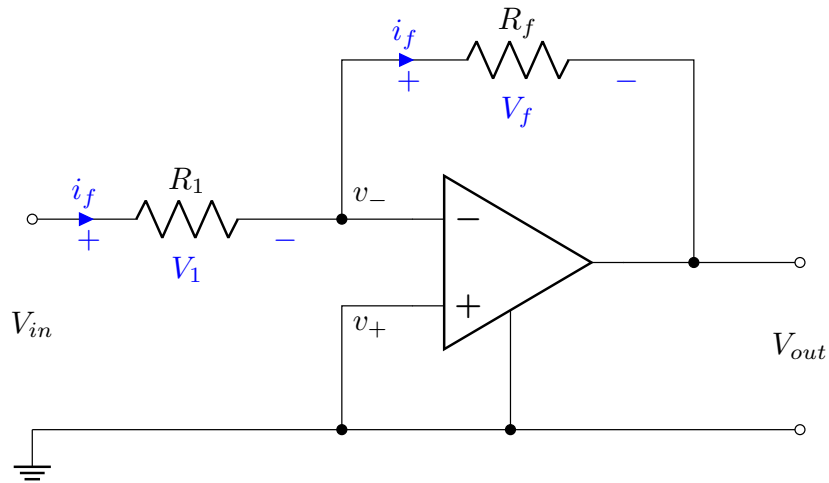


Figure 2: Inverting amplifier.

Start with the signals and parameters:

Symbol	Units	Description
v_{in}	[V]	Input voltage
v_{out}	[V]	Output voltage
v_1	[V]	Voltage across R_1
v_f	[V]	Voltage across R_f
i_f	[A]	Current through R_1 and R_f

Table 1: Symbols of Problem 2.A

Parameter	Units	Description
R_1	$[\Omega(\frac{V}{A})]$	Resistance
R_f	$[\Omega(\frac{V}{A})]$	Resistance

Table 2: Parameters of Problem 2.A

Symbols:

$$v_{in}(t) = v_{out}(t) + v_f(t) + v_1(t)$$

$$v_-(t) = v_{in} - v_1(t)$$

$$v_-(t) = v_{out}(t) + v_f(t)$$

$$v_1(t) = R_1 i_f(t)$$

$$v_f(t) = R_f i_f(t)$$

Introduce parameters:

$$v_{in} = v_{out} + R_f i_f + R_1 i_f$$

$$v_- = v_{in} - R_1 i_f$$

Transform:

$$V_{in}(s) = V_{out}(s) + (R_f + R_1)I_f(s)$$

$$V_-(s) = V_{in}(s) - R_1 I_f(s)$$

$$V_{out}(s) = -\frac{10^7}{s+1}V_-(s)$$

Substitute:

$$V_- = -10^{-7}(s+1)V_{out}$$

$$-10^{-7}(s+1)V_{out} = V_{in} - R_1 I_f$$

$$R_1 I_f = V_{in} + 10^{-7}(s+1)V_{out}$$

$$I_f = \frac{1}{R_1}(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} = V_{out} + (R_f + R_1)\frac{1}{R_1}(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} = V_{out} + \left(\frac{R_f}{R_1} + 1\right)(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} - \left(\frac{R_f}{R_1} + 1\right)V_{in} = V_{out} + \left(\frac{R_f}{R_1} + 1\right)10^{-7}(s+1)V_{out}$$

$$R_1 V_{in} - (R_f + R_1)V_{in} = R_1 V_{out} + (R_f + R_1)10^{-7}(s+1)V_{out}$$

$$-R_f V_{in} = (R_1 + 10^{-7}(R_f + R_1)(s+1))V_{out}$$

$$G_{amp} = \frac{V_{out}}{V_{in}} = \frac{-R_f}{R_1 + 10^{-7}(R_f + R_1)(s+1)}$$

$$G_{amp} = \frac{V_{out}}{V_{in}} = \frac{-R_f}{10^{-7}(R_f + R_1)s + 10^{-7}(R_f + R_1) + R_1}$$

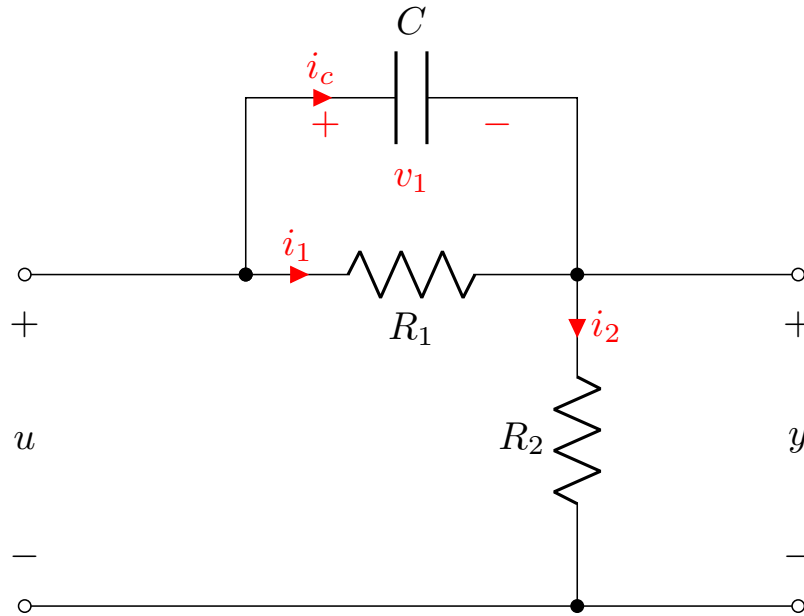
Problem 2.B.a — Passive lead circuit

Figure 3: Passive lead circuit.

Symbol	Units	Description
$u(t)$	[V]	Input voltage
$y(t)$	[V]	Output voltage
$v_1(t)$	[V]	Voltage across R_1 and C
$i_1(t)$	[A]	Current through R_1
$i_c(t)$	[A]	Current through C
$i_2(t)$	[A]	Current through R_2

Table 3: Symbols of Problem 2.B.a

Parameter	Units	Description
R_1	$[\Omega(\frac{V}{A})]$	Resistance
R_2	$[\Omega(\frac{V}{A})]$	Resistance
C	$[F(\frac{A \cdot sec}{V})]$	Capacitance

Table 4: Parameters of Problem 2.B.a

Symbol equations:

$$u(t) = y(t) + v_1(t)$$

$$i_2(t) = i_1(t) + i_C(t)$$

Parameter equations:

$$i_c(t) = C\dot{v}_1(t)$$

$$i_2(t) = \frac{y(t)}{R_2}$$

$$i_1(t) = \frac{v_1(t)}{R_1}$$

Substitute:

$$\frac{y}{R_2} = \frac{v_1}{R_1} + C\dot{v}_1$$

$$R_1 y = R_2 v_1 + R_1 R_2 C \dot{v}_1$$

Frequency Domain:

$$U(s) = Y(s) + V_1(s)$$

$$R_1 Y(s) = (R_1 R_2 C s + R_2) V_1(s)$$

$$(R_1 R_2 C s + R_2) V_1(s) = R_1 Y(s)$$

$$V_1(s) = \frac{R_1}{R_1 R_2 C s + R_2} Y(s)$$

$$U(s) = Y(s) + \frac{R_1}{R_1 R_2 C s + R_2} Y(s)$$

$$U(s) = \left(\frac{R_1 R_2 C s + R_2}{R_1 R_2 C s + R_2} + \frac{R_1}{R_1 R_2 C s + R_2} \right) Y(s)$$

$$U(s) = \frac{R_1 R_2 C s + R_2 + R_1}{R_1 R_2 C s + R_2} Y(s)$$

$$G_{yu}(s) = \frac{Y(s)}{U(s)} = \frac{R_1 R_2 C s + R_2}{R_1 R_2 C s + R_2 + R_1}$$

Units:

$$\frac{\frac{V}{A} \frac{V}{A} \frac{A \cdot \text{sec}}{V} \text{sec}^{-1} + \frac{V}{A} + \frac{V}{A}}{\frac{V}{A} \frac{V}{A} \frac{A \cdot \text{sec}}{V} \text{sec}^{-1} + \frac{V}{A}}$$

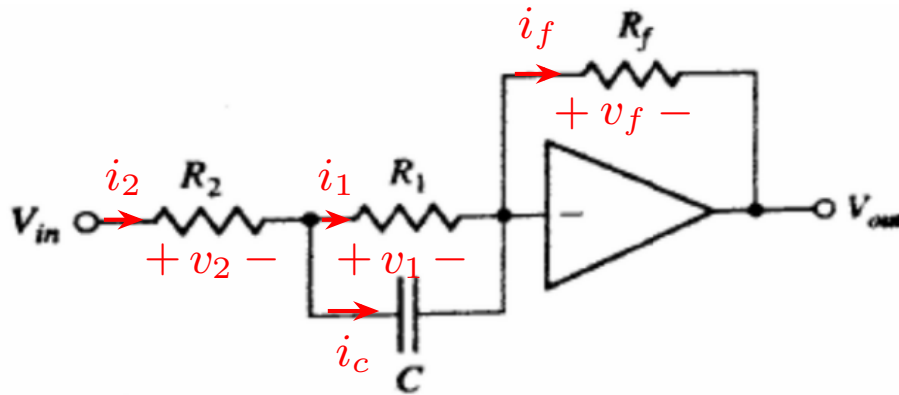
Problem 2.B.b — Active Lead Circuit


Figure 4: Active lead circuit.

Symbol	Units	Description
$v_{in}(t)$	[V]	Input voltage
$v_{out}(t)$	[V]	Output voltage
$v_1(t)$	[V]	Voltage across R_1 and C
$v_2(t)$	[V]	Voltage across R_2
$v_f(t)$	[V]	Voltage across R_f
$i_2(t)$	[A]	Current through R_2
$i_c(t)$	[A]	Current through C
$i_1(t)$	[A]	Current through R_1
$i_f(t)$	[A]	Current through R_f

Table 5: Symbols of Problem 2.B.b

Parameter	Units	Description
R_1	$[\Omega(\frac{V}{A})]$	Resistance
R_2	$[\Omega(\frac{V}{A})]$	Resistance
R_f	$[\Omega(\frac{V}{A})]$	Resistance
C	$[F(\frac{A \cdot sec}{V})]$	Capacitance

Table 6: Parameters of Problem 2.B.b

Symbol equations:

$$\begin{aligned}i_f &= i_2 = i_1 + i_c \\v_{in} &= v_2 + v_1 + v_f + v_{out} \\v_{in} &= v_2 + v_1 \\v_{out} &= -v_f\end{aligned}$$

Parameter equations:

$$\begin{aligned}i_2 &= \frac{v_2}{R_2} \\i_1 &= \frac{v_1}{R_1} \\i_c &= C\dot{v}_1 \\i_2 &= \frac{v_f}{R_f}\end{aligned}$$

Start combining:

$$\begin{aligned}v_f &= -v_{out} \\v_2 &= \frac{R_2}{R_f}v_f = -\frac{R_2}{R_f}v_{out} \\v_{in} &= -\frac{R_2}{R_f}v_{out} + v_1\end{aligned}$$

$$\begin{aligned}\frac{1}{R_f}v_f &= -\frac{1}{R_f}v_{out} = \frac{v_1}{R_1} + C\dot{v}_1 \\ \frac{1}{R_f}v_{out} &= -\frac{1}{R_1}v_1 - C\dot{v}_1\end{aligned}$$

Transform:

$$\begin{aligned}V_{in}(s) &= -\frac{R_2}{R_f}V_{out}(s) + V_1(s) \\ \frac{1}{R_f}V_{out}(s) &= -(Cs + \frac{1}{R_1})V_1(s)\end{aligned}$$

Combine and substitute:

$$\begin{aligned}-(R_1R_fCs + R_f)V_1(s) &= R_1V_{out} \\ V_1(s) &= -\frac{R_1}{R_f(R_1Cs + 1)}V_{out}(s)\end{aligned}$$

$$\begin{aligned}V_{in}(s) &= -\frac{R_2}{R_f}V_{out}(s) + V_1(s) \\ V_{in}(s) &= -\frac{R_2}{R_f}V_{out}(s) - \frac{R_1}{R_f(R_1Cs + 1)}V_{out}(s) \\ V_{in}(s) &= -(\frac{R_2}{R_f} + \frac{R_1}{R_f(R_1Cs + 1)})V_{out}(s) \\ V_{in}(s) &= -\frac{R_2(R_1Cs + 1) + R_1}{R_f(R_1Cs + 1)}V_{out}(s)\end{aligned}$$

$$G_b = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_f(R_1Cs + 1)}{R_2(R_1Cs + 1) + R_1}$$

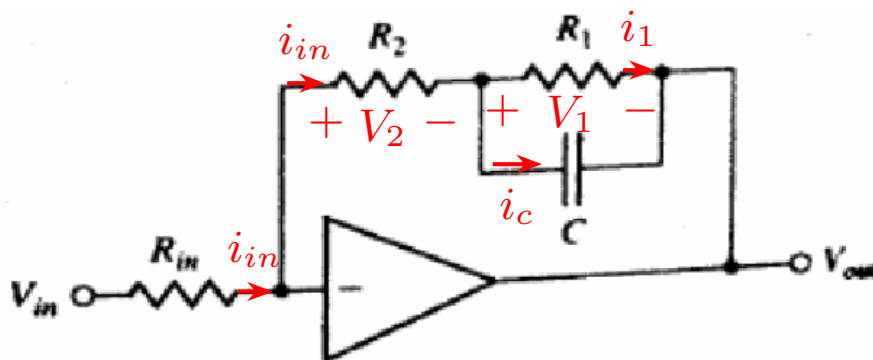
Problem 2.B.c — Active lag circuit


Figure 5: Active lag circuit.

Symbol	Units	Description
$v_{in}(t)$	[V]	Input voltage
$v_{out}(t)$	[V]	Output voltage
$v_2(t)$	[V]	Voltage across R_2
$v_1(t)$	[V]	Voltage across R_1 and C
$i_{in}(t)$	[A]	Current through R_{in} and R_2
$i_c(t)$	[A]	Current through C
$i_1(t)$	[A]	Current through R_1

Table 7: Symbols of Problem 2.B.c

Parameter	Units	Description
R_{in}	$[\Omega(\frac{V}{A})]$	Resistance
R_1	$[\Omega(\frac{V}{A})]$	Resistance
R_2	$[\Omega(\frac{V}{A})]$	Resistance
C	$[F(\frac{A \cdot sec}{V})]$	Capacitance

Table 8: Parameters of Problem 2.B.c

$$i_{in}(t) = i_1(t) + i_c(t)$$

$$v_{out}(t) = -v_1(t) - v_2(t)$$

$$i_{in}(t) = \frac{1}{R_{in}}v_{in}(t)$$

$$i_{in}(t) = \frac{1}{R_2}v_2(t)$$

$$v_{in}(t) = R_{in}i_{in}(t)$$

$$i_1(t) = \frac{1}{R_1}v_1(t)$$

$$i_c(t) = C\dot{v}_1(t)$$

Get rid of v_2 :

$$\frac{1}{R_2}v_2(t) = \frac{1}{R_{in}}v_{in}(t)$$

$$v_2(t) = \frac{R_2}{R_{in}}v_{in}(t)$$

Get rid of $i_{in}(t)$:

$$i_{in}(t) = \frac{1}{R_2}v_2(t)$$

$$i_{in}(t) = \frac{1}{R_2} \frac{R_2}{R_{in}}v_{in}(t)$$

$$i_{in}(t) = \frac{1}{R_{in}}v_{in}(t)$$

Get the first equation to standard form:

$$i_{in}(t) = i_1(t) + i_c(t)$$

$$\frac{1}{R_{in}}v_{in}(t) = \frac{1}{R_1}v_1(t) + C\dot{v}_1(t)$$

$$R_1v_{in}(t) = R_{in}(v_1(t) + R_1C\dot{v}_1(t))$$

And the second:

$$v_{out}(t) = -v_1(t) - v_2(t)$$

$$v_{out}(t) = -v_1(t) - \frac{R_2}{R_{in}}v_{in}(t)$$

$$R_{in}v_{out}(t) = -R_{in}v_1(t) - R_2v_{in}(t)$$

Frequency Domain:

$$\begin{aligned}R_{in}V_{out}(s) &= -R_{in}V_1(s) - R_2V_{in}(s) \\ R_1V_{in}(s) &= R_{in}(R_1Cs + 1)V_1(s)\end{aligned}$$

Solve for $V_1(s)$:

$$\begin{aligned}R_{in}(R_1Cs + 1)V_1(s) &= R_1V_{in}(s) \\ V_1(s) &= \frac{R_1}{R_{in}(R_1Cs + 1)}V_{in}(s)\end{aligned}$$

Plug it back into the first Freq domain question:

$$\begin{aligned}R_{in}V_{out}(s) &= -R_{in}V_1(s) - R_2V_{in}(s) \\ R_{in}V_{out}(s) &= -R_{in}\frac{R_1}{R_{in}(R_1Cs + 1)}V_{in}(s) - R_2V_{in}(s) \\ R_{in}V_{out}(s) &= -\left(\frac{R_1}{R_1Cs + 1} + R_2\right)V_{in}(s) \\ R_{in}(R_1Cs + 1)V_{out}(s) &= -(R_1 + R_2(R_1Cs + 1))V_{in}(s)\end{aligned}$$

Finally, the Transfer function:

$$G_c = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{R_1 + R_2(R_1Cs + 1)}{R_{in}(R_1Cs + 1)}$$

Problem 3 — Read Head

Symbol	Units	Description
$\theta_1(t)$	$[\frac{rad}{sec}]$	Angular position of motor
$\theta_2(t)$	$[\frac{rad}{sec}]$	Angular position of read head
$M_C(t)$	$[N \cdot m]$	Command torque (moment)
$M_D(t)$	$[N \cdot m]$	Disturbance torque (e.g. friction)

Table 9: Symbols of Problem 2.C

Parameter	Units	Description
I_1	$[\frac{N \cdot m \cdot sec^2}{rad}]$	rotational inertia of motor
I_2	$[\frac{N \cdot m \cdot sec^2}{rad}]$	rotational inertia of head
b	$[\frac{N \cdot m \cdot sec}{rad}]$	damping coefficient of flexible shaft
k	$[\frac{N \cdot m}{rad}]$	spring coefficient of flexible shaft

Table 10: Parameters of Problem 2.C

Characteristic equations (given by problem):

$$I_1 \ddot{\theta}_1 + b(\dot{\theta}_1 - \dot{\theta}_2) + k(\theta_1 - \theta_2) = M_C + M_D$$

$$I_2 \ddot{\theta}_2 + b(\dot{\theta}_2 - \dot{\theta}_1) + k(\theta_2 - \theta_1) = 0$$

Rearrange:

$$I_2 \ddot{\theta}_2 + b\dot{\theta}_2 + k\theta_2 = b\dot{\theta}_1 + k\theta_1$$

$$M_C + M_D = I_1 \ddot{\theta}_1 + b\dot{\theta}_1 + k\theta_1 - b\dot{\theta}_2 - k\theta_2$$

Frequency Domain:

$$(I_2 s^2 + bs + k)\Theta_2(s) = (bs + k)\Theta_1(s)$$

$$\Theta_1(s) = \frac{I_2 s^2 + bs + k}{bs + k} \Theta_2(s)$$

Frequency Domain of second equation:

$$M_C(s) = (I_1 s^2 + bs + k)\Theta_1(s) - (bs + k)\Theta_2(s)$$

$$M_C(s) = (I_1 s^2 + bs + k) \frac{I_2 s^2 + bs + k}{bs + k} \Theta_2(s) - (bs + k)\Theta_2(s)$$

$$M_C(s) = \frac{(I_1 s^2 + bs + k)(I_2 s^2 + bs + k)}{bs + k} \Theta_2(s) - \frac{(bs + k)^2}{bs + k} \Theta_2(s)$$

$$M_C(s) = \frac{(I_1 s^2 + bs + k)(I_2 s^2 + bs + k) - (bs + k)^2}{bs + k} \Theta_2(s)$$

Collect Terms:

s^4	s^3	s^2	s^1	s^0
$I_1 I_2$	$I_1 b$	$I_1 k$		
	$I_2 b$	b^2	bk	
		$I_2 k$	bk	k^2
		$-b^2$	$-2bk$	$-k^2$

Combine and produce output:

$$G_C(s) = \frac{\Theta_2(s)}{M_C(s)} = \frac{bs + k}{I_1 I_2 s^4 + (I_1 + I_2)bs^3 + (I_1 + I_2)ks^2}$$